

Taxes and Transfers

Economic Impact of government Measures



Edoardo Cefala



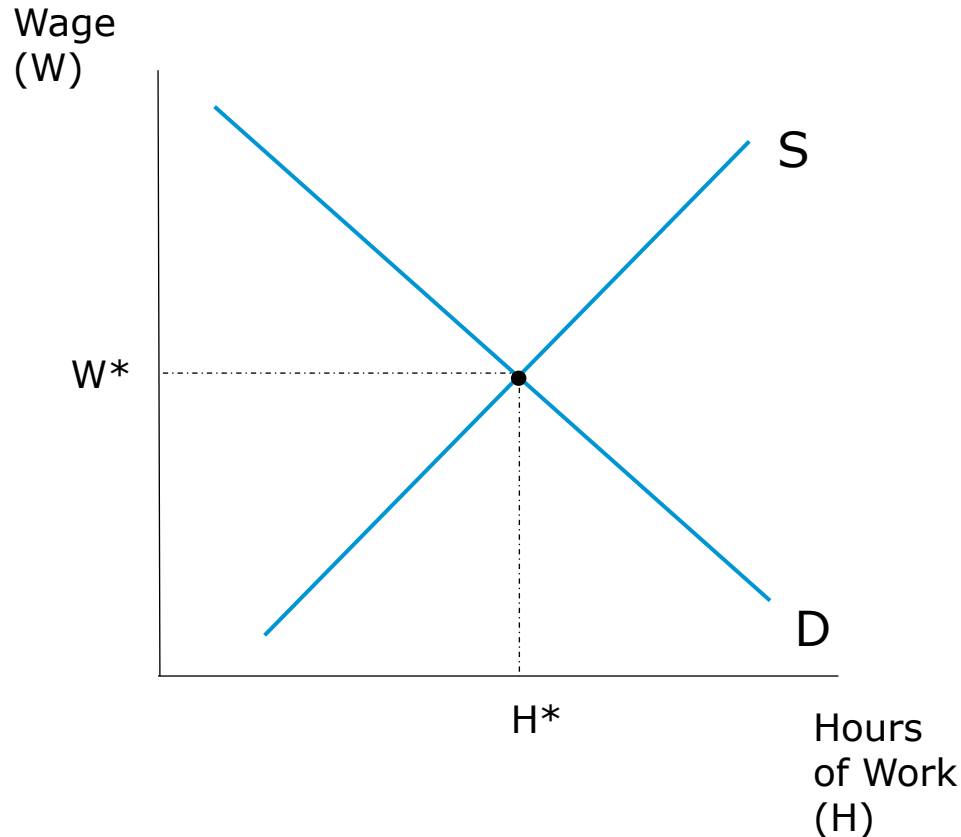
Taxes and Transfers in the Labor Factor Market. Aggregate.



Chapter 20 (4)

- Typical Factor Markets: Labor and Capital
- Analogous to Tax Incidence in Goods Markets
- Difference: Individuals now supply the factor of labor (Producers)
- Firms demand the factor of labor (Consumers)
- The price is now the wage
- The quantity is the hours of work

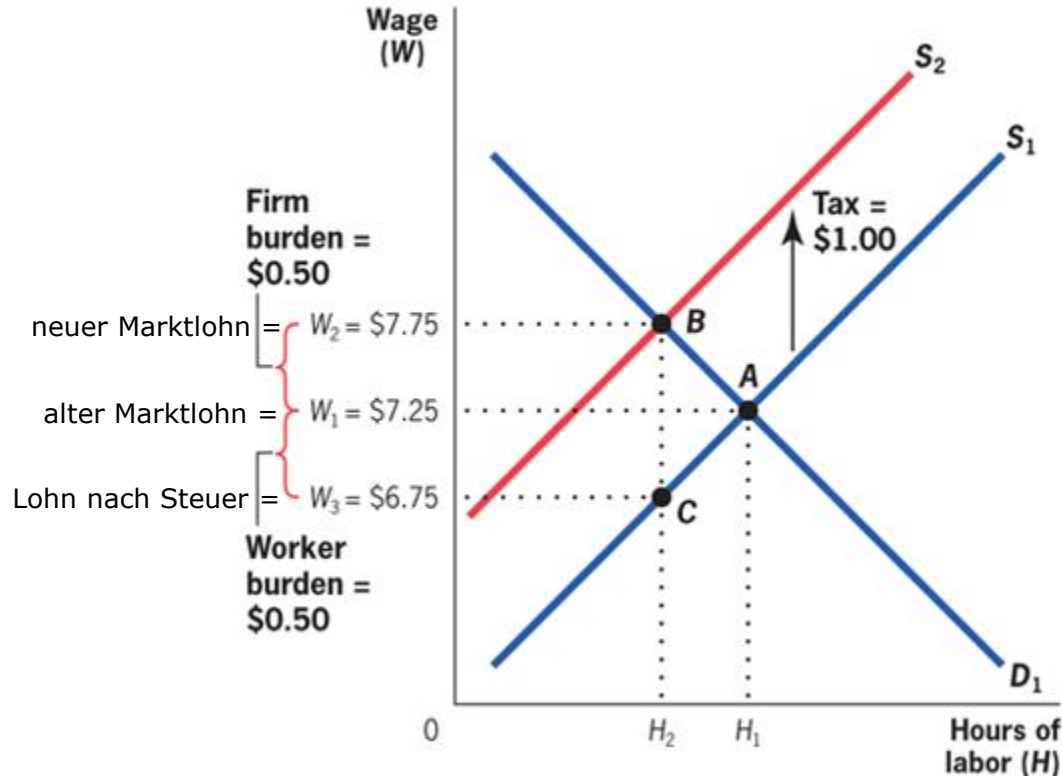
Factor Markets II



- S = Labor Supply Curve
- Hours of work that employees are willing to offer at a specific wage
- D = Labor Demand Curve
- Hours of work that employers demand at a specific wage
- A = Intersection of Supply and Demand determines the market equilibrium

Impact of Tax and Factor Market I

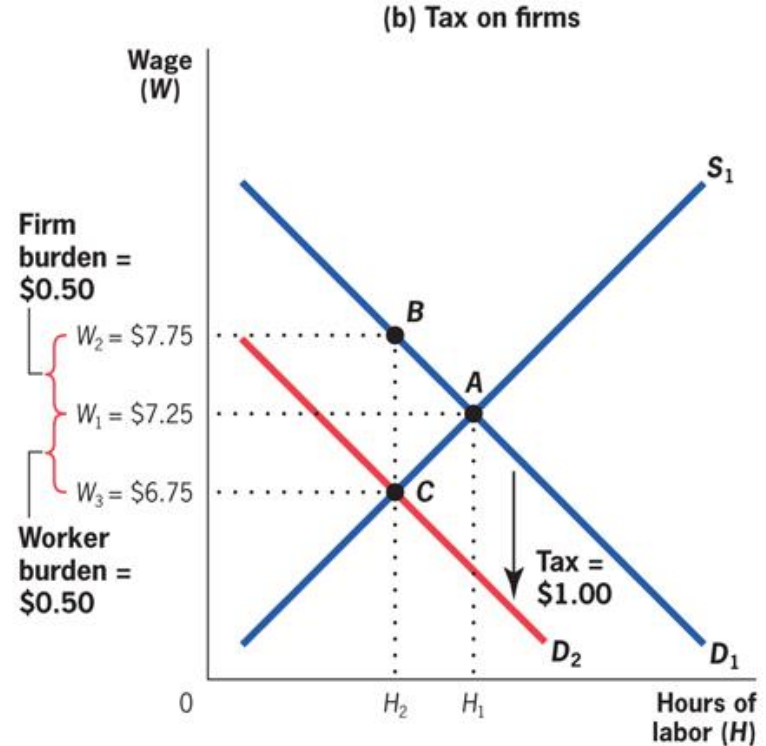
(a) Tax on workers



- **Tax imposed on employees**
- Tax on labor of $\$1 \rightarrow$ for every hour of work, employees get $\$1$ less
- For every level of labor supply, a wage of $\$1$ more is now demanded (Shift of the supply curve upwards)
- Less labor is supplied at a higher equilibrium wage
- The burden is borne by employers and employees

Impact of Tax and Factor Market II

- **Tax imposed on companies/employers**
- Companies must pay \$1 as tax -> for every hour of work, companies are now only willing to pay \$1 less
- For every level of labor supply, a wage of \$1 less is offered (Shift of the demand curve downwards)
- Less labor is supplied at a lower equilibrium wage
- The burden is borne by employers and employees



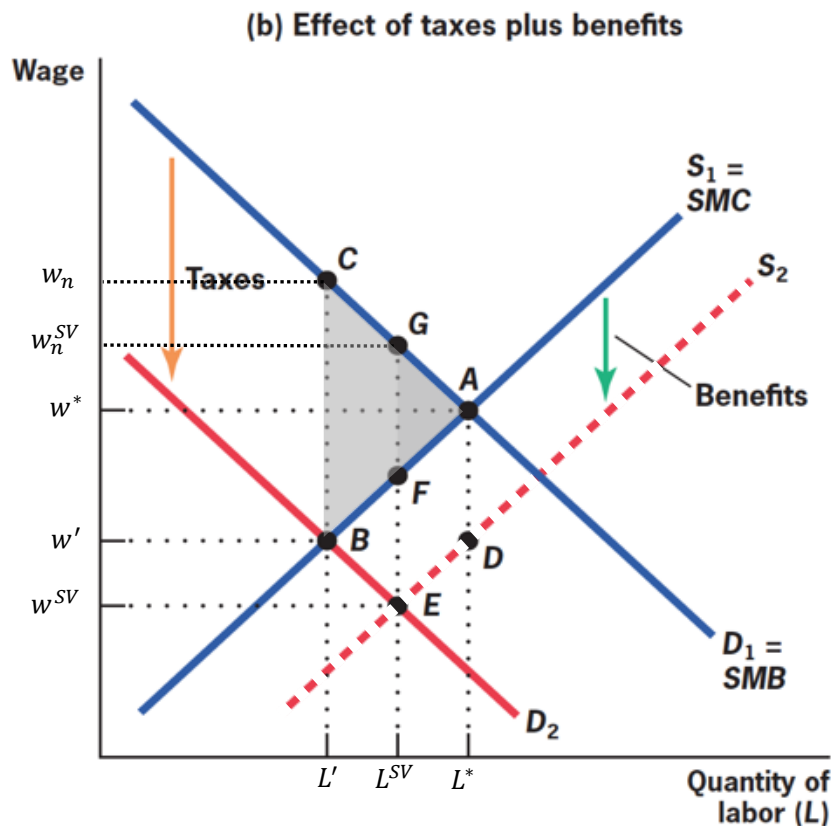
Impact of Tax and Factor Market III

- **New market wage** ("new intersection point") = the wage that is paid/received by the side that **DOES NOT** have to pay the tax to the state
- **Wage after tax** = the wage that is paid/received by the side that must pay the tax to the state
- **Employers pay the tax:** $\text{Wage after tax} = \text{Market wage} + \text{Tax}$
- **Employees pay the tax:** $\text{Wage after tax (= income)} = \text{Market wage} - \text{Tax}$

- Connection between tax revenues and government expenditures
- Relationship between taxes and transfer payments
- Effects on tax incidence?
- Effects on efficiency?

Financing of Transfers

A simple model

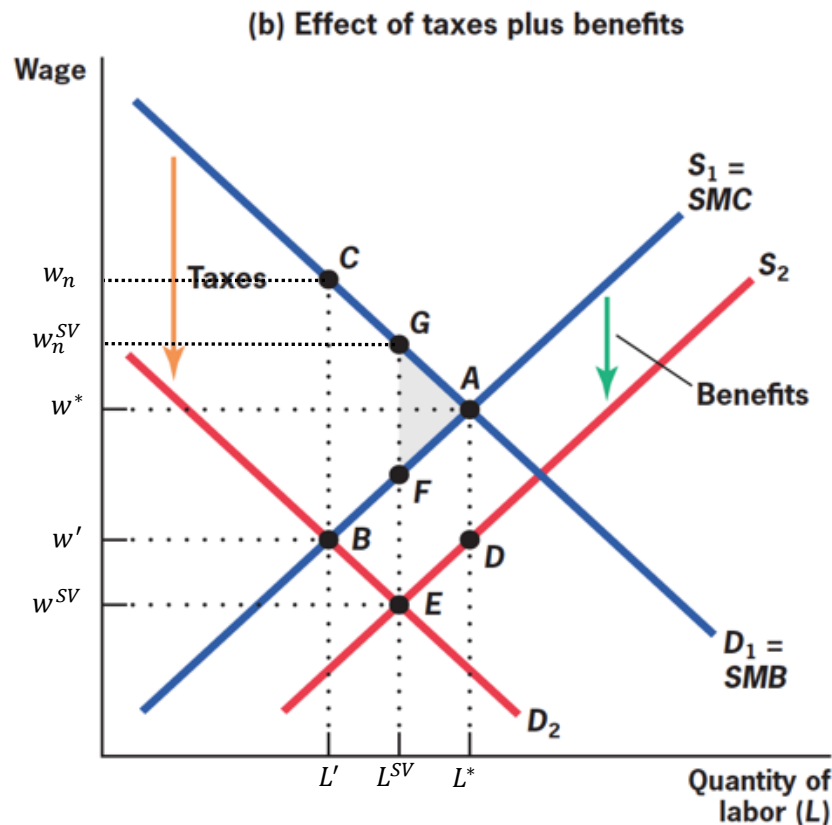


- **Introduction of a tax** paid by the employers. This increases the costs for the employers and shifts the demand curve downwards.
- Equilibrium from $A \rightarrow B$
- The wage sinks from $w^* \rightarrow w'$ (new market wage, which the employees receive)
- $w_n = w' + \text{tax} = \text{wage after tax, which the employers pay}$
- The quantity of labor decreases ($L^* > L'$)
- A welfare loss occurs (ΔCAB)

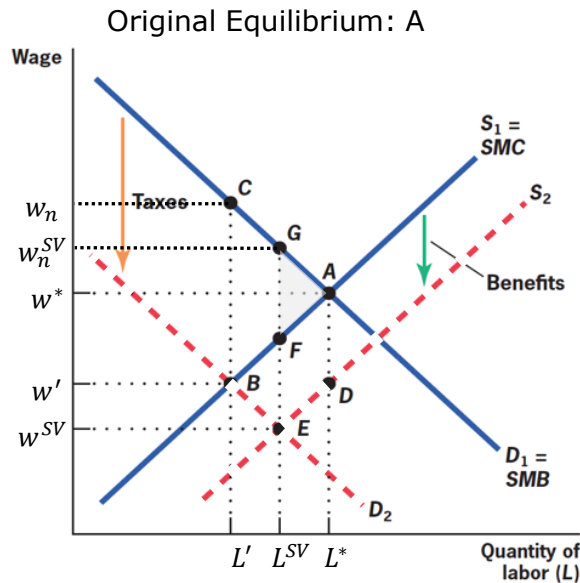
Financing of Transfers

A simple model

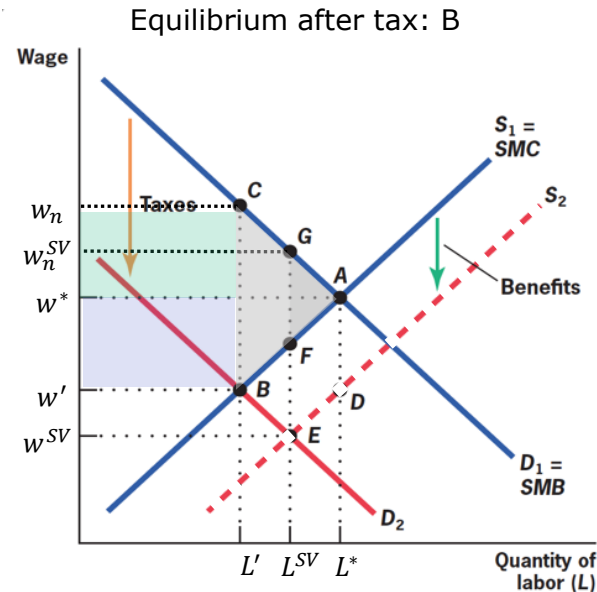
- The revenues from the tax are used to finance transfers -> a benefit arises for employees
- Employees are willing to work for a lower wage because they benefit from transfers
 - "Employees do not have to spend a portion of their income to purchase, for example, social security, kindergarten, or a place at a university on the market"
- Labor supply shifts downwards
- Equilibrium from B -> E
- The wage sinks ($w' \rightarrow w^{SV}$)
- The quantity of labor increases ($L' \rightarrow L^{SV}$)
- The welfare loss decreases ($\Delta CAB \rightarrow \Delta AGF$)



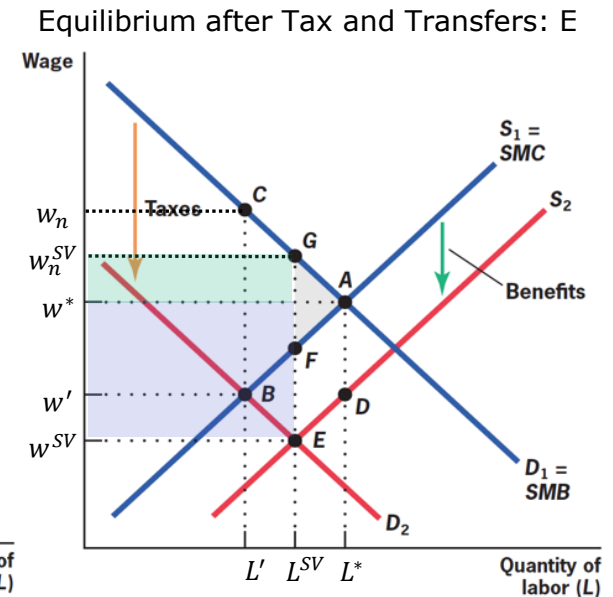
Tax Burden Distribution & Welfare Loss



No Tax Burden & Welfare Loss



Tax Burden : AG, AN
Welfare Loss : ΔCAB

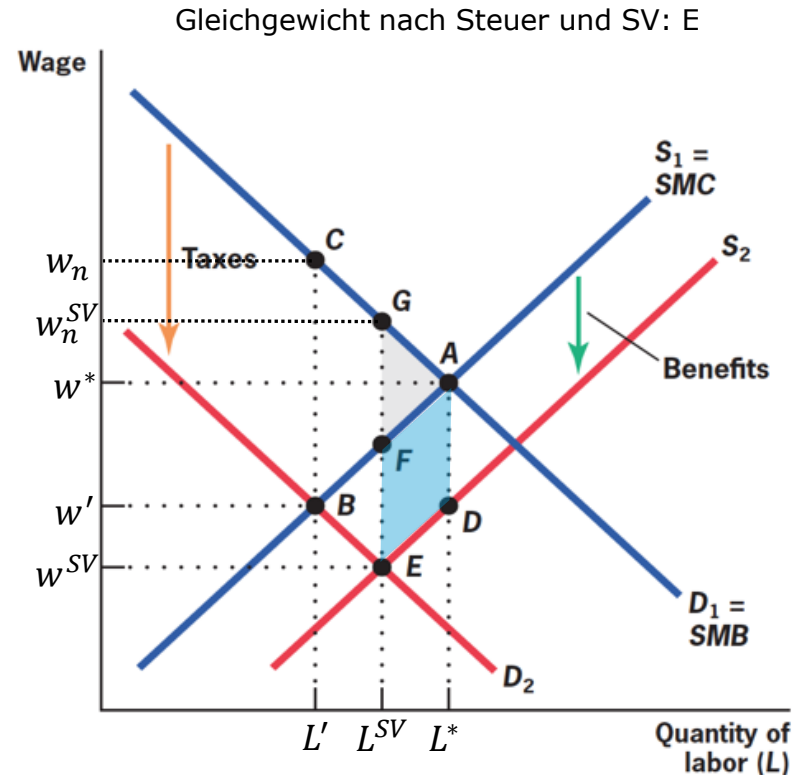


Tax Burden : AG, AN
Welfare Loss : ΔAGF

Financing of Transfers

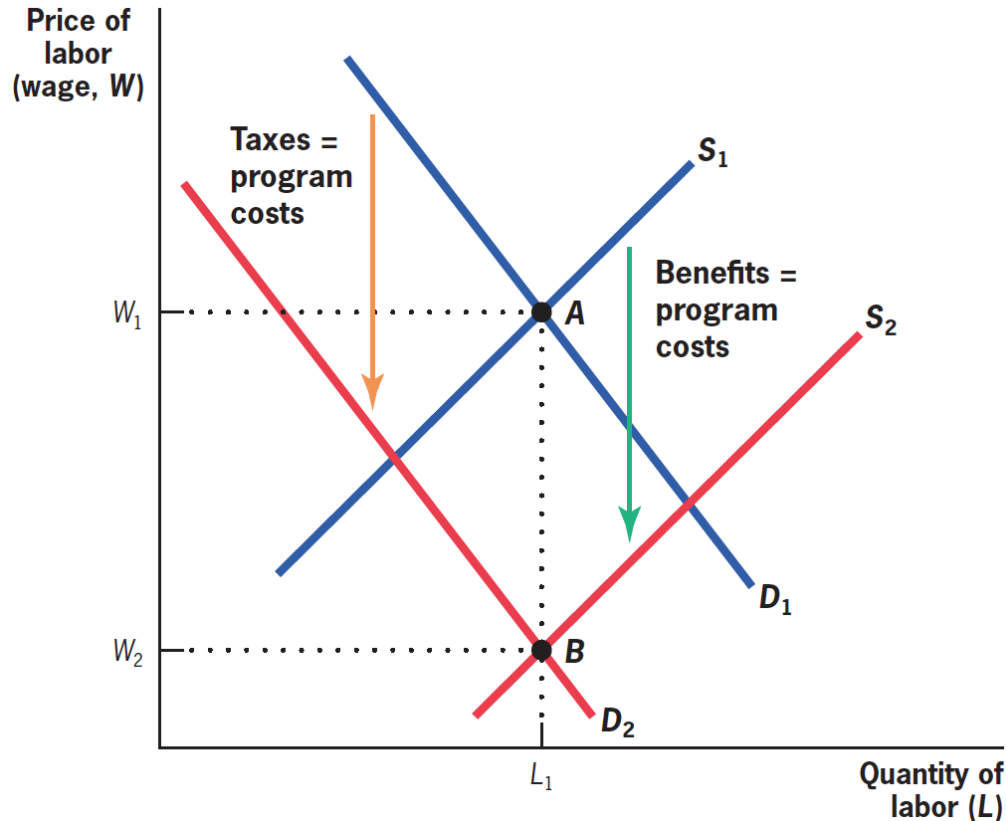
Welfare Loss with Transfers

- Equilibrium after Tax and Social Security: E
- Why doesn't the rectangle AFED count as welfare loss?
- Welfare loss always refers to the original market equilibrium and thus the original supply and demand functions!
- Rent under S 1 would also not have been generated (and therefore cannot be "lost")



Financing of Transfers

Extreme Case: Benefit to Employees = Cost of the Tax



- If the benefit to employees from transfers equals the cost of the tax, there is no welfare loss!
- The employees bear the full tax burden! ($W_1 \rightarrow W_2$)

Quelle: Gruber, 5th edition, S. 646

Financing of Transfers

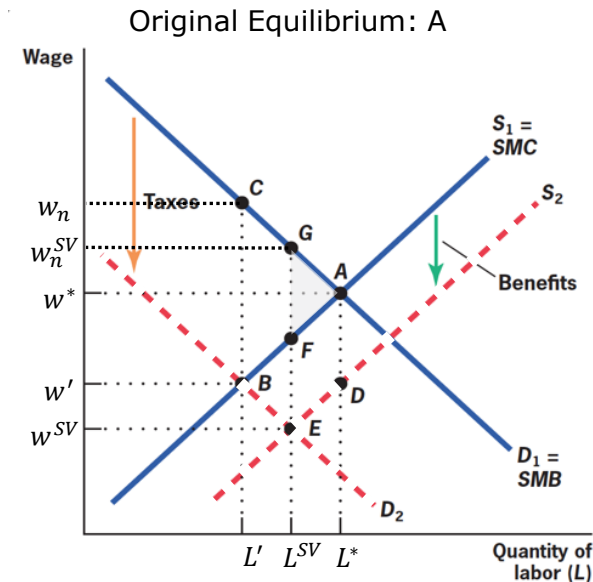
Example

- Assume that the demand function in a labor market is $w = 1000 - 2L$, and the supply function is $w = 3L$ (where w ... wage, L ... labor in time).
- Labor is taxed with a quantity tax (per hour) of 100€. This is paid by the employers.
- **Calculate the equilibrium wage and the corresponding quantity (L) in the market without tax.**
- **Calculate the market wage and the number of hours of work in equilibrium with the tax.**
- **Assume that the revenue from the quantity tax is used to finance a social security for the employees. Employees reduce the wage they demand by 50 cents for every euro of tax revenue. Calculate the market wage and the number of hours of work.**
- **Calculate the distribution of the tax burden, as well as the welfare loss with and without considering social security.**

Financing of Transfers

Example

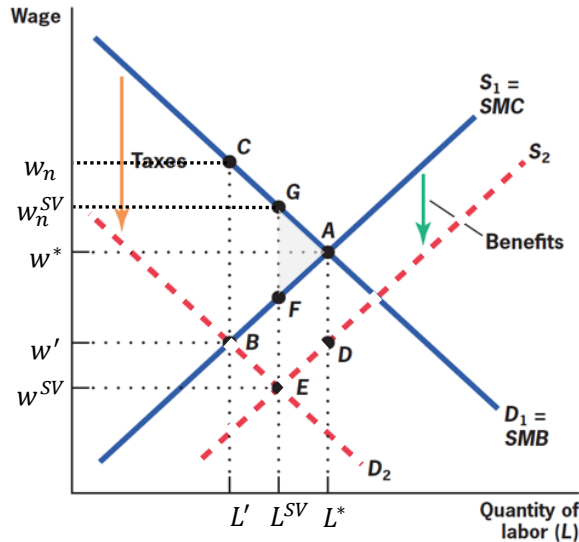
- Assume that the demand function in a labor market is $w = 1000 - 2L$, and the supply function is $w = 3L$ (where w ... wage, L ... labor in time).
- Labor is taxed with a quantity tax (per hour) of 100€. This is paid by the employers.
- Calculate the equilibrium wage and the corresponding quantity (L) in the market without tax.**



Financing of Transfers

Example

Ursprüngliches Gleichgewicht: A



$$1000 - 2L = 3L$$

$$L^* = 200$$

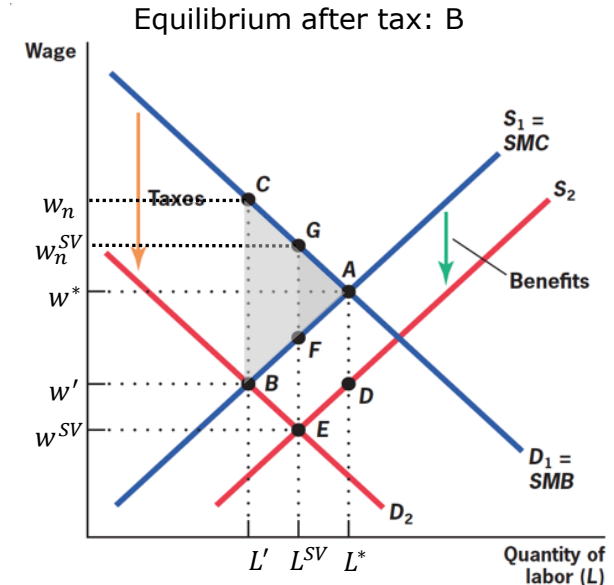
$$w = 3L \rightarrow w^* = 600$$

Financing of Transfers

Example

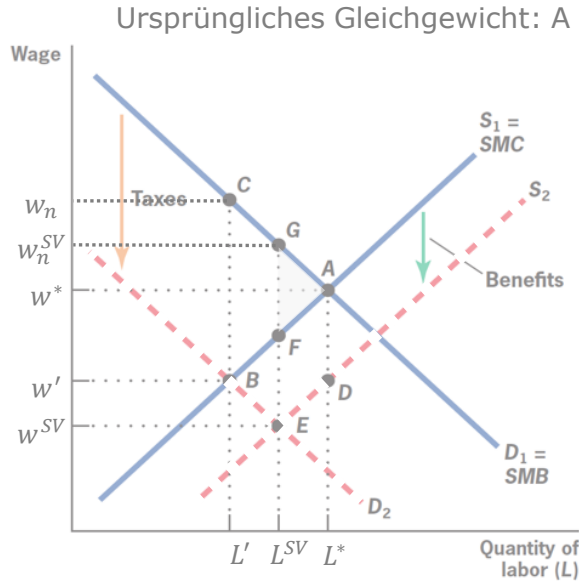
- Assume that the demand function in a labor market is $w = 1000 - 2L$, and the supply function is $w = 3L$ (where w ... wage, L ... labor in time).
- Labor is taxed with a quantity tax (per unit/hour) of 100€. This is paid by the employers.
- Calculate the market wage as well as the number of hours of work in equilibrium with tax.**

→ New Demand Function: $w = 1000 - 2L - 100$



Financing of Transfers

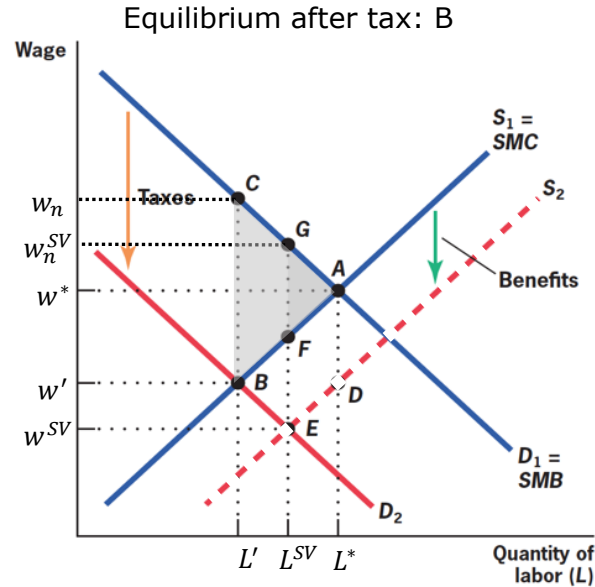
Example



$$1000 - 2L = 3L$$

$$L^* = 200$$

$$w = 3L \rightarrow w^* = 600$$



$$1000 - 2L - 100 = 3L$$

$$L' = 180,$$

$$w' = 3 * 180 = 540$$

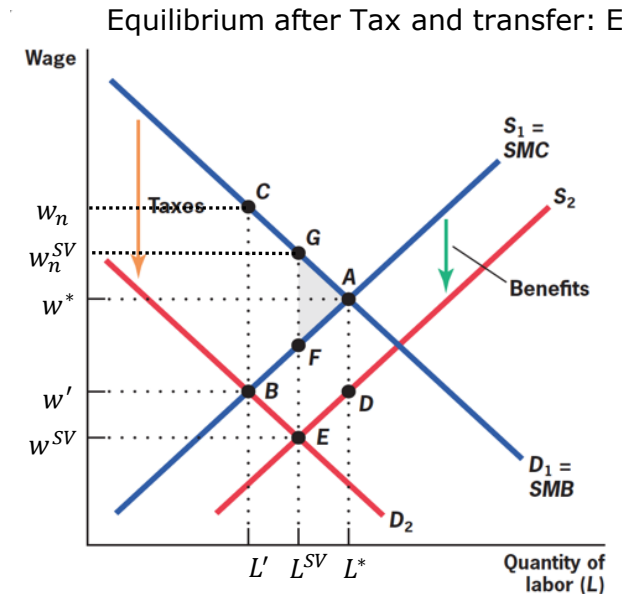
$$w_n = 1000 - 2 * 180 = 640$$

Tax paid by
employers

Financing of Transfers

Example

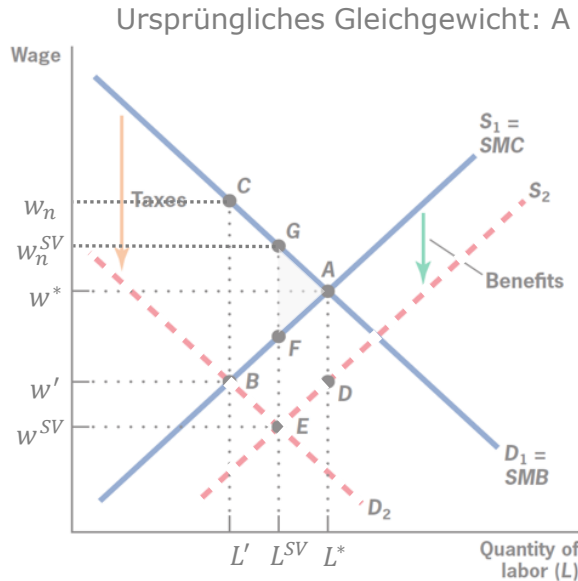
- Assume that the demand function in a labor market is $w = 1000 - 2L$, and the supply function is $w = 3L$ (where w ... wage, L ... labor in time).
- Labor is taxed with a quantity tax (per unit/hour) of 100€. This is paid by the employers.
- Assume that the revenue from the quantity tax is used to finance a social security for the employees. Employees reduce the wage they demand by 50 cents for every euro of tax revenue. Calculate the market wage and the number of hours of work.**



→ New Supply function: $w = 3L - 100 * 0,5$

Financing of Transfers

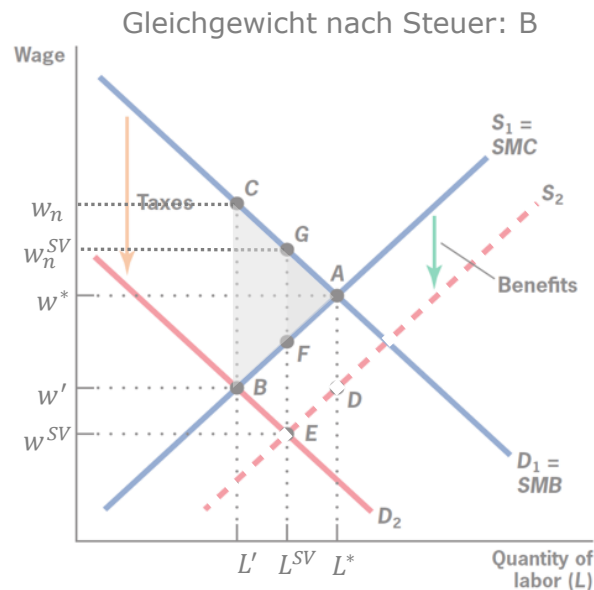
Example



$$1000 - 2L = 3L$$

$$L^* = 200$$

$$w = 3L \rightarrow w^* = 600$$

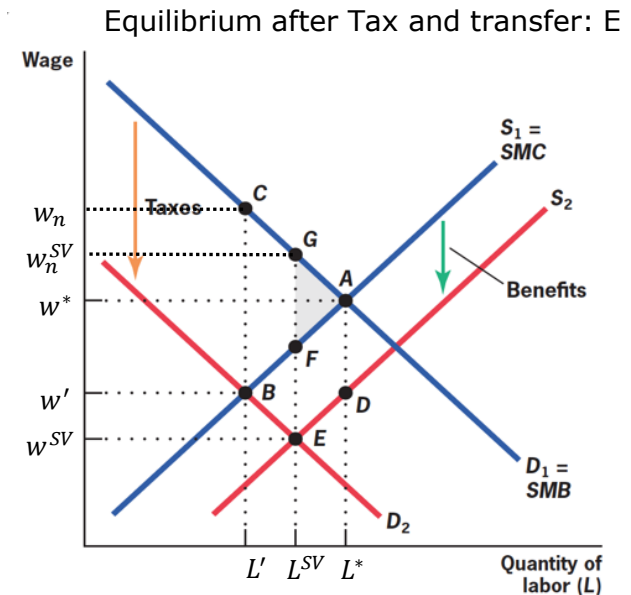


$$1000 - 2L - 100 = 3L$$

$$L' = 180,$$

$$w' = 3 * 180 = 540$$

$$w_n = 1000 - 2 * 180 = 640$$



$$1000 - 2L - 100 = 3L - 100 * 0,5$$

$$L^{SV} = 190,$$

$$w^{SV} = 3 * 190 - 50 = 520$$

$$w_n^{SV} = 1000 - 2 * 190 = 620$$

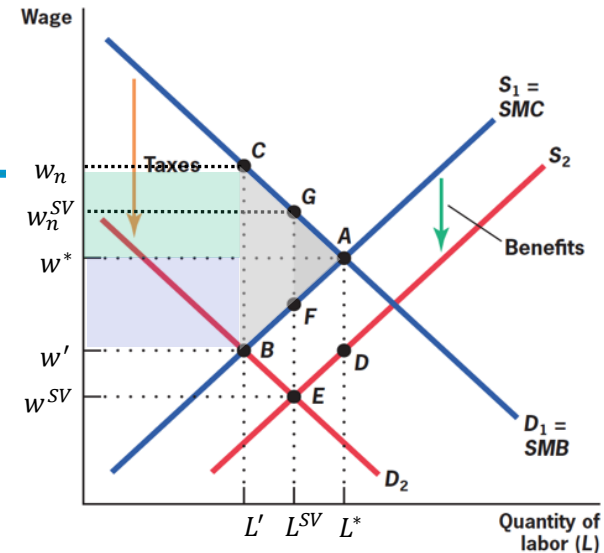
Employees are willing to work for a lower wage because they receive benefits from transfers (here, social security).

Financing of Transfers

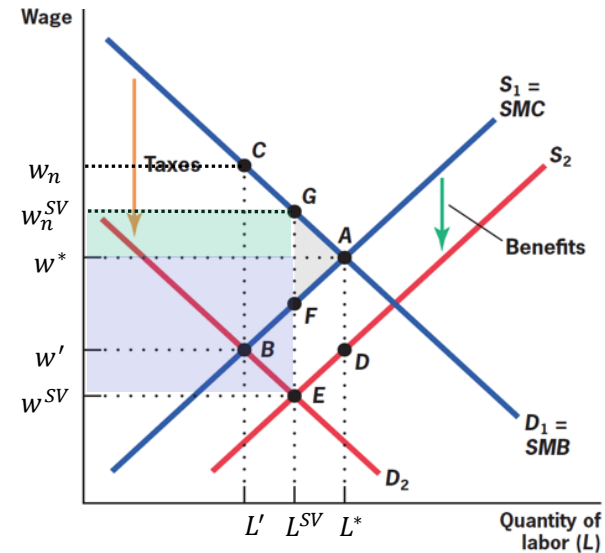
Example

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- Labor is taxed with a quantity tax (per hour) of 100€. This is paid by the employers.
- Calculate the distribution of the tax burden, as well as the welfare loss with and without considering social security.**

Gleichgewicht nach Steuer: B



Gleichgewicht nach Steuer und Transfers: E

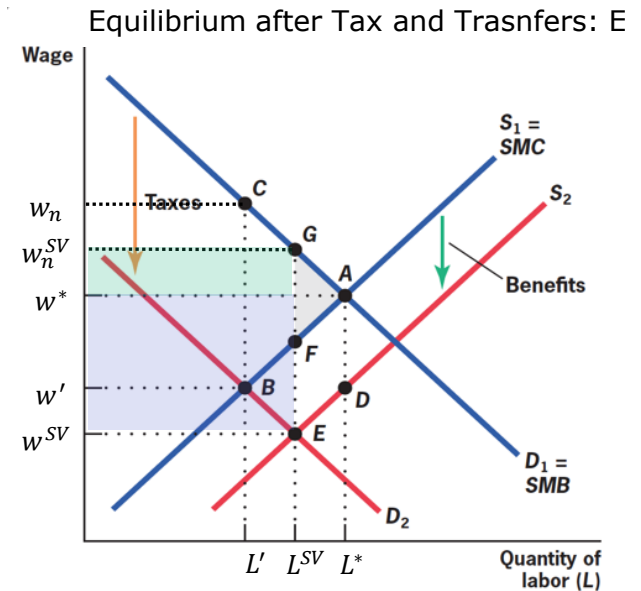
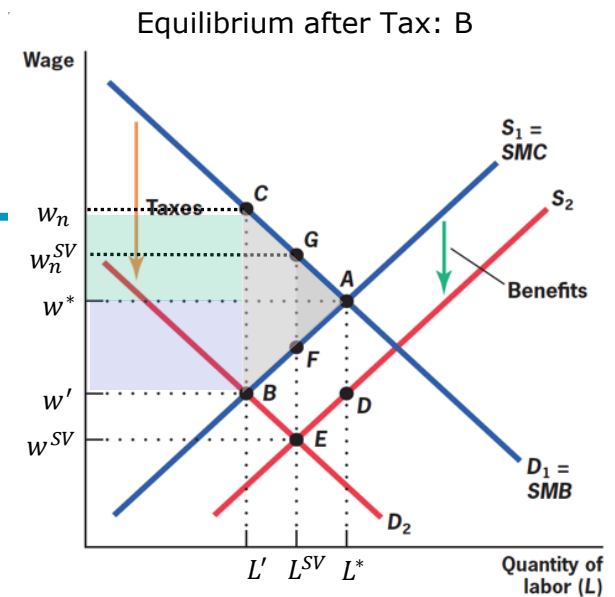


Finanzierung von Transfers

Solution

- **Tax Burden Distribution**
- **Without Transfers in Equilibrium B:**
- **Employees:** $w^* - w' = 600 - 540 = 60 \rightarrow 60\%$
- **Employers:** $w_n - w^* = 640 - 600 = 40 \rightarrow 40\%$
- **With Transfers in Equilibrium E:**
- **Employees:** $w^* - w^{SV} = 600 - 520 = 80 \rightarrow 80\%$
- **Employers:** $w_n^{SV} - w^* = 620 - 600 = 20 \rightarrow 20\%$

With transfers, employees bear a larger part of the tax burden.



Finanzierung von Transfers

Solution

- Welfare Loss**

- Without Transfers in Equilibrium B:**

- Employees (ABH):**

$$2(w^* - w') * (L^* - L') = 2(600 - 540) * (200 - 180) = 600 \rightarrow 60\%$$

- Employers (ACH):**

$$2(w_n - w^*) * (L^* - L') = 2(640 - 600) * (200 - 180) = 400 \rightarrow 40\%$$

- With Transfers in Equilibrium E:**

- Employees (AFI):**

$$2(w^* - w_x) * (L^* - L^{SV}) = 2(600 - 570) * (200 - 190) = 150 \rightarrow 60\%$$

- Employers (AGI):**

$$2(w_n^{SV} - w^*) * (L^* - L^{SV}) = 2(620 - 600) * (200 - 190) = 100 \rightarrow 40\%$$

Neue Menge in alte Angebotsfunktion

Mit Sozialversicherung sinkt der Wohlfahrtsverlust, die Verteilung bleibt aber gleich.

Wohlfahr Gleichgewicht nach Steuer: B

Ohne Transfers in Gleichgewicht B:

- Arbeitnehmer:innen (ABH): $\frac{(w^* - w') * (L^* - L')}{2} = \frac{(600 - 540) * (200 - 180)}{2} = 600 \rightarrow 60\%$

- Arbeitgeber:innen (ACH): $\frac{(w_n - w^*) * (L^* - L')}{2} = \frac{(640 - 600) * (200 - 180)}{2} = 400 \rightarrow 40\%$

Transfers in Gleichgewicht E:

- Arbeitnehmer:innen (AFI): $\frac{(w^* - w_x) * (L^* - L^{SV})}{2} = \frac{(600 - 570) * (200 - 190)}{2} = 150 \rightarrow 60\%$

- Arbeitgeber:innen (AGI): $\frac{(w_n^{SV} - w^*) * (L^* - L^{SV})}{2} = \frac{(620 - 600) * (200 - 190)}{2} = 100 \rightarrow 40\%$

Wohl Gleichgewicht nach Steuer und Transfers: E

Ohne Transfers in Gleichgewicht B:

- Arbeitnehmer:innen (ABH): $\frac{(w^* - w') * (L^* - L')}{2} = \frac{(600 - 540) * (200 - 180)}{2} = 600 \rightarrow 60\%$

- Arbeitgeber:innen (ACH): $\frac{(w_n - w^*) * (L^* - L')}{2} = \frac{(640 - 600) * (200 - 180)}{2} = 400 \rightarrow 40\%$

Transfers in Gleichgewicht E:

- Arbeitnehmer:innen (AFI): $\frac{(w^* - w_x) * (L^* - L^{SV})}{2} = \frac{(600 - 570) * (200 - 190)}{2} = 150 \rightarrow 60\%$

- Arbeitgeber:innen (AGI): $\frac{(w_n^{SV} - w^*) * (L^* - L^{SV})}{2} = \frac{(620 - 600) * (200 - 190)}{2} = 100 \rightarrow 40\%$

Social Security Programs



- **Social Insurances**
- The welfare state provides social insurances for negative shocks: health, pension, accident, and unemployment insurance
- The idea of a community supporting people in emergencies (illness, unemployment, etc.) -> social values
- A large part of the expansion of the welfare state and increasing government expenditures in the 20th century is due to social insurances
- Externalities of insurance for all (especially health insurance)

- insurance markets are very susceptible to market failures:
 - Systemic: asymmetric information -> adverse selection, moral hazard

Adverse Selection: incomplete information about the "quality/risk" of those to be insured -> pushes those with "better" risk (low risk) out of the market ("lemons" problem), Problem: insurance offers protection at a cost that does not cover the risk

Moral Hazard: lack of incentive to avoid risk, - "I'm insured anyway" - because others bear the costs of the risk

- Individual behavioral errors: short-sightedness, too little information, too little self-discipline
- -> Mandatory insurance for all at the same price (compensation)

Social Security Programs in Austria

- Health, pension, accident insurance



Österreichische
Sozialversicherung

Unemployment insurance
Labor market service (AMS)



- Contributions are paid in roughly equal parts by employees and employers (accident insurance only by employers)
- Maximum contribution base: 6,450.00€ gross per month; social security contributions must be paid up to this income threshold, income above that remains social security contribution-free -> regressive

Financing of Social Security

Empirical Questions

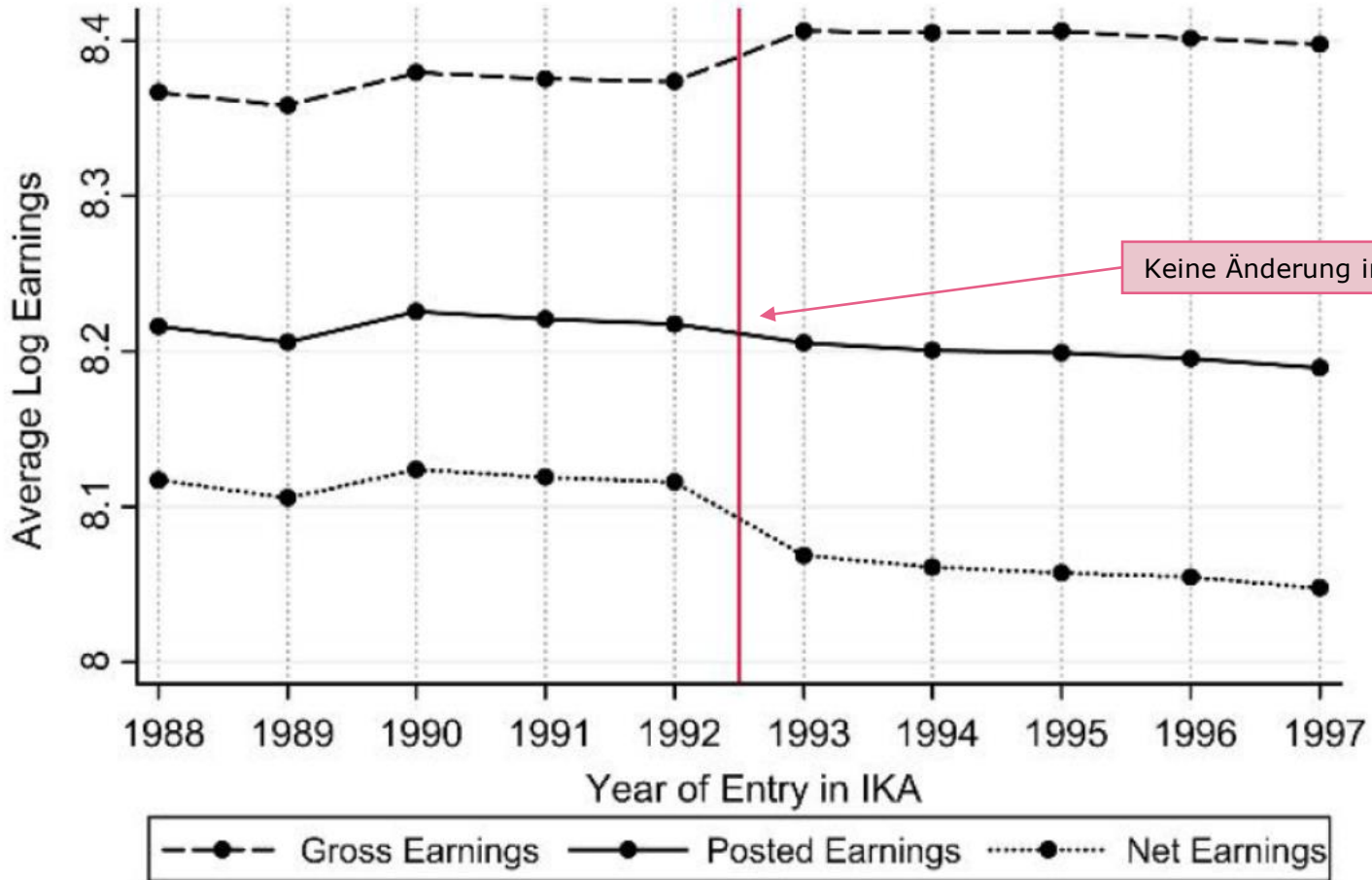
- If, in the best-case scenario, the introduction of a social insurance has no costs for the employers, why is it then introduced by the state and not by the employers?
- What does it depend on, how much benefit a social insurance brings to employees?
- What does the distribution of the tax burden depend on?

- USA 1970: In some states, employers must cover the costs for pregnancies and births.
- These costs were not previously covered by social insurance.
 - Insurance costs for women of childbearing age increase.
- **Setup**
- Comparison of wages and labor supply between states where a reform is implemented (treatment group) and those that did not introduce the reform -> Diff-in-Diff analysis
- **Results**
- The wages of female employees sink, but the quantity of labor hardly falls.
 - The costs are borne by the female employees.

- Greece 1993: For newly hired employees, the maximum contribution for social contributions (maximum assessment base) doubles.
- For all those who were employed before 01.01.1993, the old maximum contribution still applies.
- **Setup**
- Regression Discontinuity Design to find out whether the wages of employees change.
- **Results**
- Gross wages increase.
- Employers do not pass their part of the contributions on to the employees.
- Net wages for employees sink because social contributions also increase.

B

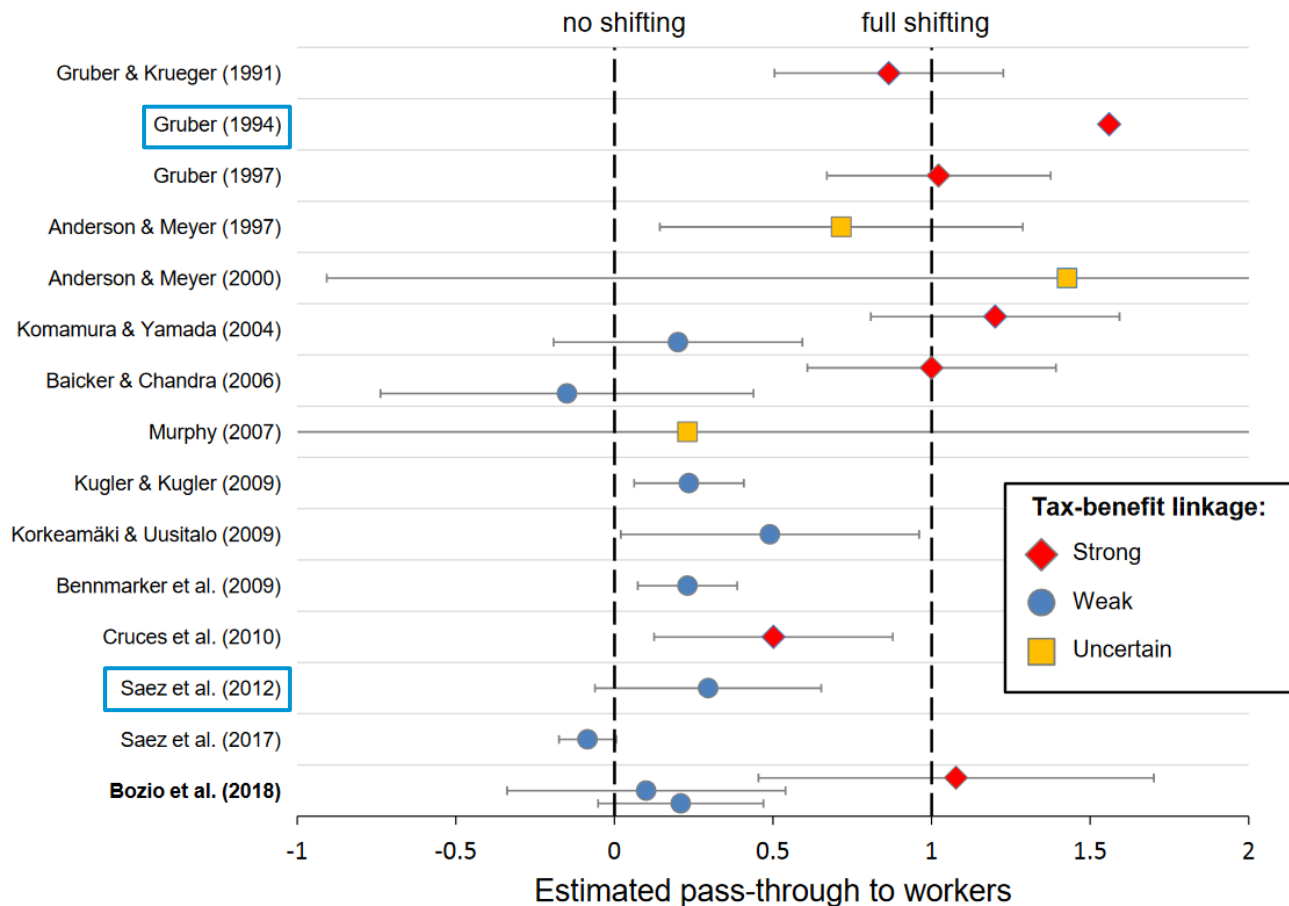
Average Log Earnings Above Old Cap



What does it depend on, how much the employees bear the financing of social security?

- The distribution of the burden depends on how strong the connection is between the payments and the benefits
- → How much does the supply curve shift as a response to the introduction of social security?
- In the first example, the connection was very clear, as the benefits only went to women of childbearing age.
- In the second example, the connection was less clear, as it was a general social security contribution that benefited everyone.

Distribution of the Tax Burden between Employers and Employees



References

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[Statistik Austria - Kindertagesheime Kinderbetreuung](#)

[Econlib - Negative Income Tax](#)

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